

HOMI BHABHA BALVAIDNYANIK COMPETITION

Class: 6 • Subject: Science • Topic: State of Matter

1. Matter and the Particle Theory

Matter is anything that has mass and occupies space. Matter is made up of extremely small particles.

Core Principles

- Particles of matter are extremely small.
- There are spaces between particles.
- Particles are continuously moving.
- Particles attract one another.
- Increasing temperature generally increases the kinetic energy and movement of particles.

Conceptual link: The observable properties of solids, liquids and gases can be explained by the spacing, movement and attractive forces between their particles.

2. Solids

A solid has a fixed shape and a fixed volume because its particles are closely packed and held by comparatively strong attractive forces.

- Particles are closely packed.
- Particles mainly vibrate about fixed positions.
- Inter-particle spaces are very small.
- Attractive forces are strong.
- Solids are generally difficult to compress.
- Heating increases particle vibration and can eventually cause melting.

3. Liquids

A liquid has a fixed volume but does not have a fixed shape.

- Particles remain relatively close together.
- Particles can move or slide past one another.
- Attractive forces are weaker than in solids.
- Liquids can flow.
- Liquids are difficult to compress because the particles are still relatively close.

Exam distinction: A liquid changes its shape according to the container, but its quantity/volume remains essentially unchanged under ordinary conditions.

4. Gases

A gas has neither a fixed shape nor a fixed volume.

- Particles are widely separated.
- Particles move rapidly and randomly.
- Attractive forces are very weak.
- Gases are highly compressible.
- A gas expands to occupy the available space.
- Moving gas particles collide with container walls and produce gas pressure.

5. Comparison of States

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Property	Solid	Liquid	Gas
Shape	Fixed	Not fixed	Not fixed
Volume	Fixed	Fixed	Not fixed
Particle spacing	Very small	Small	Large
Particle movement	Vibration about fixed positions	Movement/flow	Rapid random motion
Attraction	Strong	Moderate	Very weak
Compressibility	Very low	Very low	High
Ability to flow	No	Yes	Yes

6. Change of State

A change of state is a physical change in which a substance changes from one physical state to another due mainly to changes in temperature and/or pressure. The chemical identity of the substance remains unchanged.

Process	Change	Meaning
Melting / Fusion	Solid → Liquid	Solid absorbs heat and becomes liquid
Freezing / Solidification	Liquid → Solid	Liquid loses heat and becomes solid
Vaporisation	Liquid → Gas	Liquid changes to vapour
Condensation	Gas → Liquid	Gas loses heat and becomes liquid
Sublimation	Solid → Gas	Direct solid-to-gas change
Deposition	Gas → Solid	Direct gas-to-solid change

Melting and Freezing

The temperature at which a solid changes into a liquid is its melting point. For pure ice at normal atmospheric pressure, this is 0°C.

The reverse process is freezing. At normal atmospheric pressure, pure water freezes at 0°C.

Boiling

Boiling is rapid vaporisation occurring throughout a liquid at its boiling point. For water at normal atmospheric pressure, the boiling point is 100°C.

Boiling produces vapour bubbles throughout the liquid, whereas evaporation occurs only at the surface.

Evaporation

Evaporation is the slow conversion of a liquid into vapour from its surface. It can occur at temperatures below the boiling point.

- Higher temperature → faster evaporation.
- Larger exposed surface area → faster evaporation.
- Higher wind speed → faster evaporation.
- Lower humidity → faster evaporation.

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Why cooling occurs: Higher-energy particles escape from the liquid surface during evaporation. The average energy of the remaining particles decreases, producing cooling.

Condensation

Condensation is the conversion of vapour/gas into liquid, usually due to cooling. Water droplets forming on the outside of a cold glass are an everyday example.

Sublimation and Deposition

In sublimation, a solid changes directly into gas without passing through the liquid state. Examples include camphor, naphthalene and dry ice.

The reverse direct change, gas → solid, is called deposition.

7. Evaporation vs Boiling — High-Value Comparison

Evaporation	Boiling
Surface phenomenon	Occurs throughout the liquid
Can occur below the boiling point	Occurs at the boiling point for a given pressure
Usually slow	Rapid
No bubbles throughout the liquid	Bubbles form throughout the liquid
Produces cooling	Requires continuous heat supply

8. Effect of Temperature on Particles

When matter is heated, particles gain kinetic energy and generally move faster. When matter is cooled, particle movement generally decreases.

Heating: more particle energy → greater movement → possible change of state.

Cooling: less particle energy → reduced movement → possible change toward a more closely packed state.

9. Applications and Examination-Level Reasoning**Drying of Clothes**

Clothes dry faster in sunlight because increased temperature gives water particles more energy. Wind can further increase the rate by carrying away water vapour near the cloth.

Cotton Clothes in Summer

Cotton absorbs sweat. The absorbed water then evaporates. Since evaporation requires energy, heat is taken from the body and a cooling effect is produced.

Earthen Pot Cooling

Water can seep through tiny pores in an earthen pot and evaporate from the outer surface. The evaporation removes heat from the water remaining inside.

Perfume and Diffusion

Volatile perfume particles evaporate and then spread through the air. Their movement demonstrates the continuous motion of gas particles.

Compressed Gases

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Because gas particles have large spaces between them, gases can be compressed. This property is used for storing gases in cylinders and for compressed gases such as LPG and oxygen.

Liquefaction of Gases

Under suitable conditions of pressure and temperature, gases can be converted into liquids. Liquefaction makes some gases easier to store and transport.

Steam and Heat Transfer

Steam can cause severe burns because when steam condenses on a cooler surface such as skin, it releases additional latent heat as well as transferring heat due to its high temperature.

10. High-Value Concept Connections

- Why is a gas highly compressible? Because its particles are separated by large spaces.
- Why does a liquid flow but a solid generally does not? Liquid particles can move past one another, whereas solid particles are strongly held near fixed positions.
- Why does a gas fill the entire container? Its particles move freely and rapidly in all directions and have weak attraction.
- Why does evaporation cause cooling? Higher-energy surface particles escape, reducing the average energy of the remaining liquid.
- Why does increasing surface area increase evaporation? More liquid particles are exposed at the surface and can escape.
- Why does wind increase evaporation? Moving air removes vapour from above the liquid surface, allowing more particles to escape.
- Why does low humidity favour evaporation? The surrounding air contains less water vapour and can accept more vapour.
- Why is boiling different from evaporation? Boiling occurs throughout the liquid at a particular boiling temperature for a given pressure; evaporation is a surface process that can occur at lower temperatures.
- Why does pressure affect gases strongly? Gas particles have large spaces between them, so pressure can significantly reduce those spaces.

11. High-Value Facts for Homi Bhabha

- At normal atmospheric pressure, pure ice melts at 0°C and pure water boils at 100°C.
- Boiling point depends on pressure; therefore, 100°C is specifically the boiling point of water at normal atmospheric pressure.
- Evaporation is a surface phenomenon; boiling is a bulk phenomenon.
- Evaporation can occur below the boiling point.
- Sublimation skips the liquid state.
- During a change of state, the substance remains chemically the same.
- Gas pressure results from collisions of moving gas particles with the walls of the container.
- Temperature is related to the average kinetic energy of particles.
- Stronger inter-particle attraction generally makes separation of particles more difficult.
- The properties of each state arise from differences in particle spacing, movement and attractive forces.

12. Rapid Revision

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State	Particle Arrangement	Movement	Shape	Volume
Solid	Very closely packed	Vibrate about fixed positions	Fixed	Fixed
Liquid	Close, less ordered	Move past one another	Not fixed	Fixed
Gas	Widely separated	Rapid and random	Not fixed	Not fixed

Key chain: Particle spacing + particle movement + attractive force → properties of the state → behaviour during heating/cooling → change of state.